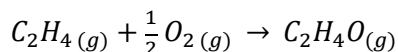


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Ethylene oxide is produced by the catalytic oxidation of ethylene:



An undesired competing reaction is the combustion of ethylene to CO₂.

The feed to a reactor contains 2 mol C₂H₄/mol O₂. The conversion and yield in the reactor are respectively 25% and 0.70 mol C₂H₄O produced/mol C₂H₄ consumed. A multiple-unit process separates the reactor outlet stream components: C₂H₄ and O₂ are recycled to the reactor, C₂H₄O is sold, and CO₂ and H₂O are discarded. The reactor inlet and outlet streams are each at 450°C, and the fresh feed and all the species leaving the separation process are at 25°C. The combined fresh feed-recycle stream is preheated to 450°C.

- Taking the basis of 2 mol of ethylene entering the reactor, draw and label a flowchart of the complete process (show the separation process as a single unit) and calculate the molar amounts of all the process streams.
- Calculate the heat requirement (kJ) for the entire process and that for the reactor alone. Use the tables below, filling in any missing entries.
- Calculate the flow rate (kg/day) of the fresh feed and the overall and reactor heat requirements (kW) for a production rate of 1500 kg C₂H₄O/day.

Useful Data:

$$\Delta H_{f,C_2H_4O}^\circ = -51.00 \frac{kJ}{mol}$$

$$\Delta H_{f,C_2H_4}^\circ = 52.28 \frac{kJ}{mol}$$

$$C_{p,C_2H_4O} [J/(mol \cdot K)] = -4.69 + 0.2061 T - 9.995 \times 10^{-5} T^2$$

$$C_{p,C_2H_4} [kJ/(mol \cdot ^\circ C)] = 40.75 \times 10^{-3} + 11.47 \times 10^{-5} T - 6.897 \times 10^{-8} T^2 + 17.66 \times 10^{-12} T^3$$

OVERALL PROCESS:

Substance	n_{in} (mol)	$\hat{H}_{in}$ (kJ/mol)	n_{out} (mol)	$\hat{H}_{out}$ (kJ/mol)
C ₂ H ₄			-	-
O ₂		0	-	-
C ₂ H ₄ O	-	-		
CO ₂	-	-		-393.5
H ₂ O(l)	-	-		-285.84

ACROSS REACTOR:

Substance	n_{in} (mol)	$\hat{H}_{in}$ (kJ/mol)	n_{out} (mol)	$\hat{H}_{out}$ (kJ/mol)
C ₂ H ₄				
O ₂		13.37		13.37
C ₂ H ₄ O	-	-		
CO ₂	-	-		-374.66
H ₂ O(g)	-	-		-226.72

